THE PECTORAL REGION

The Breast

Shape and position, the muscle bed, the lobes and ducts, the suspensory ligaments and the retromammary space, the nipple and areola, the lymph drainage, the blood supply and the clinical notes.

Dr. Khalid Milad · Anatomy Department,
Faculty of Medicine, University of Tripoli

Figures selected from Netter's Atlas of Human Anatomy, Snell's Clinical Anatomy by Regions, Moore's Clinically Oriented Anatomy and the THIEME Atlas of Anatomy, with Dr. Milad's own drawings. For private study only.

Contents

1 The Breast — Structure & Position 3

2 Lymph Drainage of the Breast 12

3 The Blood Supply of the Breast 18

4 Clinical Notes 22

CHAPTER 1

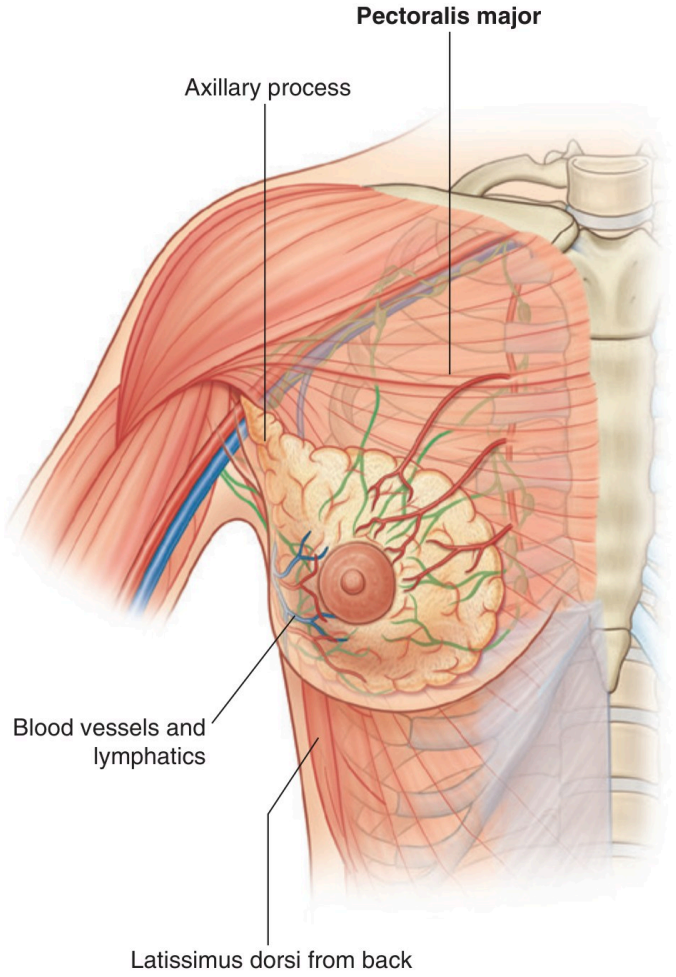
The Breast — Structure & Position

The Breast

The female breast (mamma) is formed by the mammary gland (specialized accessory gland) secreting milk.

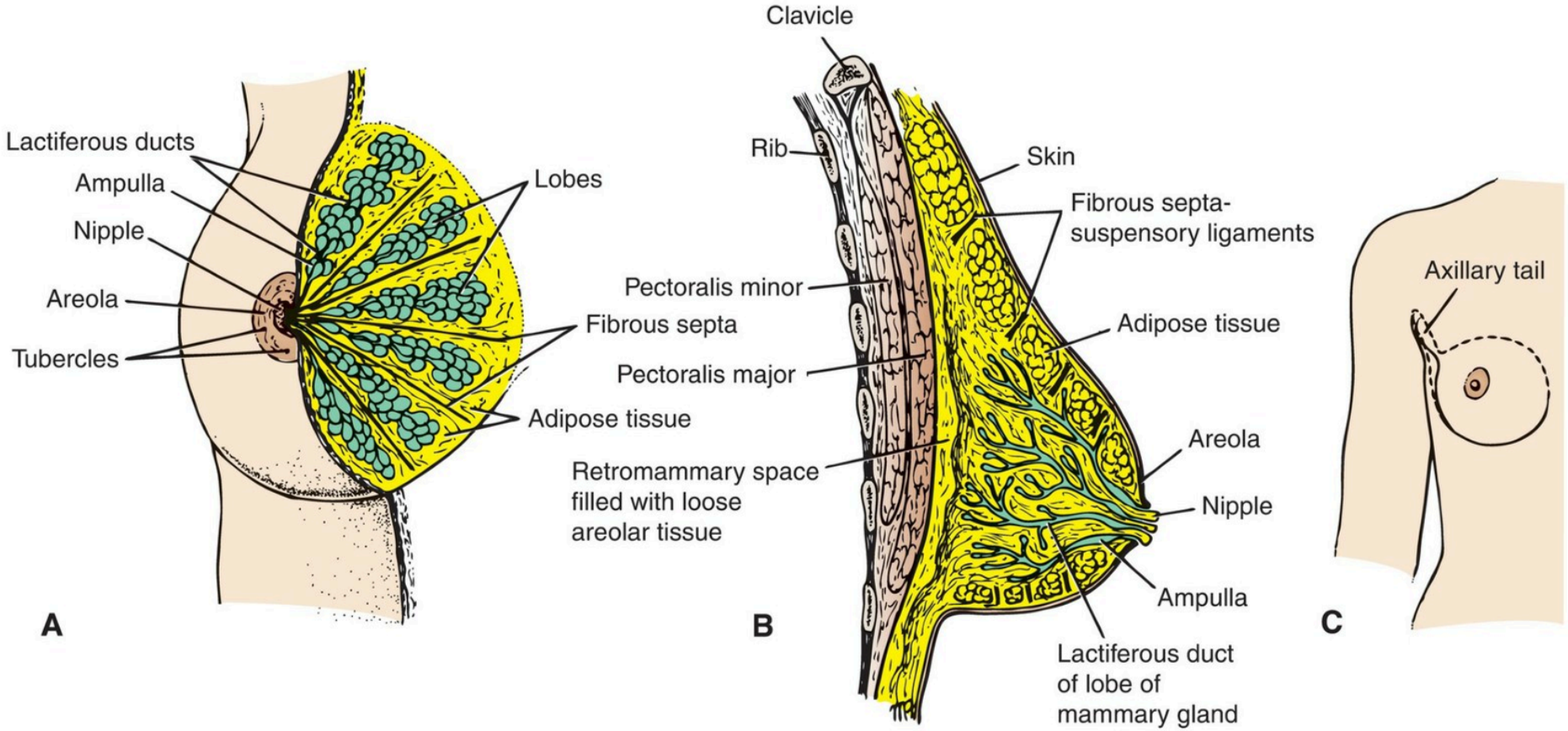
N.B.

Breast is a modified sweat (sudoriferous) gland.



The right breast in position on pectoralis major, with the axillary process running up into the axilla and the blood vessels and lymphatics of the gland.

The Breast (cont.)



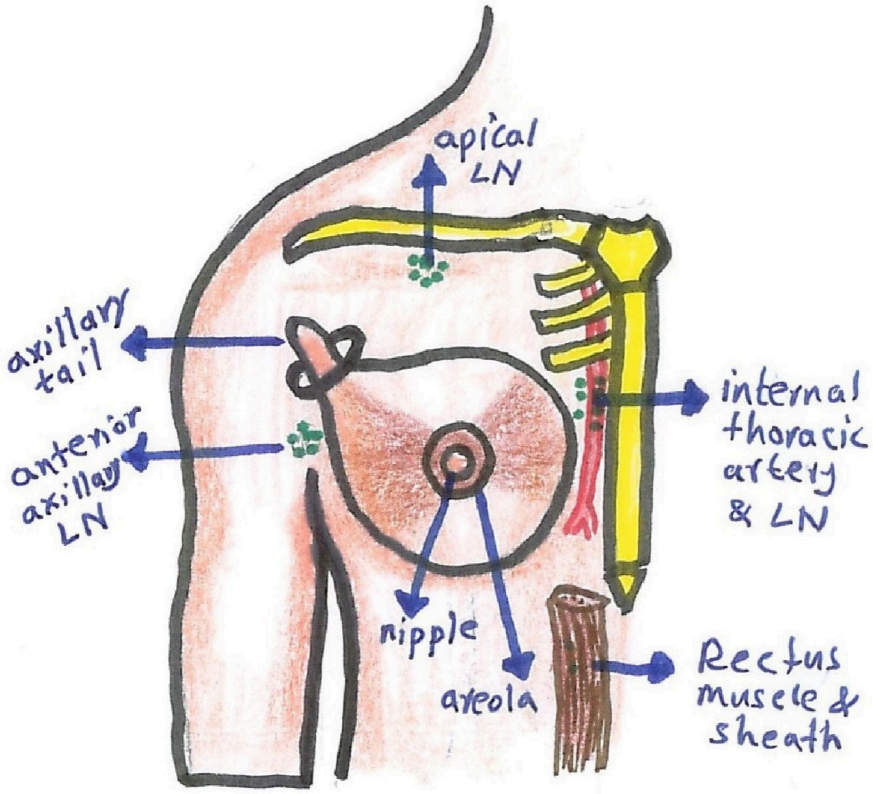
The mature female breast: A, anterior view with the skin partly removed to show the lobes, ducts and ampulla; B, sagittal section; C, the axillary tail.

Shape & Position

- The breast is conical in shape with its base extending from the second to the sixth ribs (in midclavicular line) and from lateral border of sternum to the midaxillary line.
- Present in superficial fascia, except axillary tail [pierces deep fascia at lower border of pectoralis major & enters axilla].
- Muscle bed of breast (muscles under it) are:
  - Pectoralis major (the main muscle bed).
  - Serratus anterior.
  - External oblique.

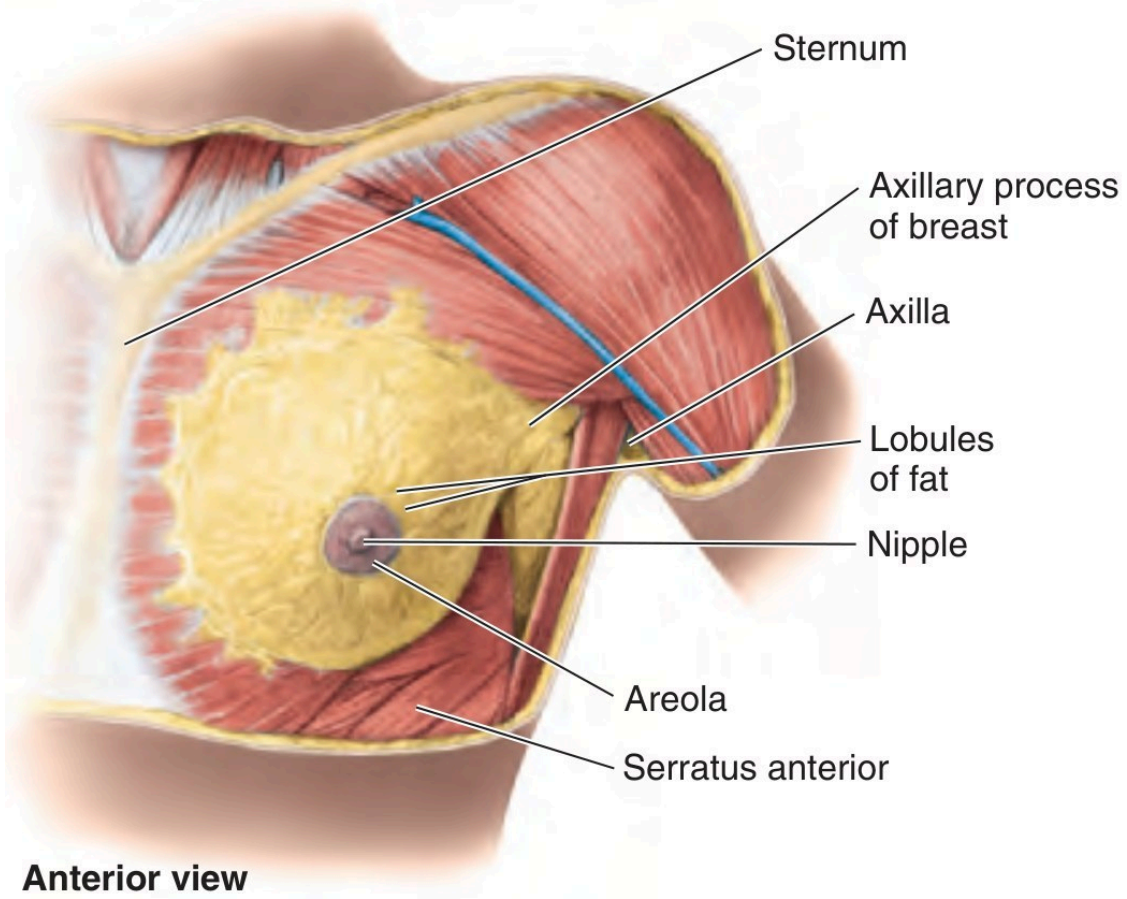
N.B.

Muscle bed: 1. Pectoralis major. 2. Serratus anterior.



Dr. Milad's own drawing: the breast in position, with the axillary tail, the anterior axillary and apical nodes, and the internal thoracic artery and nodes.

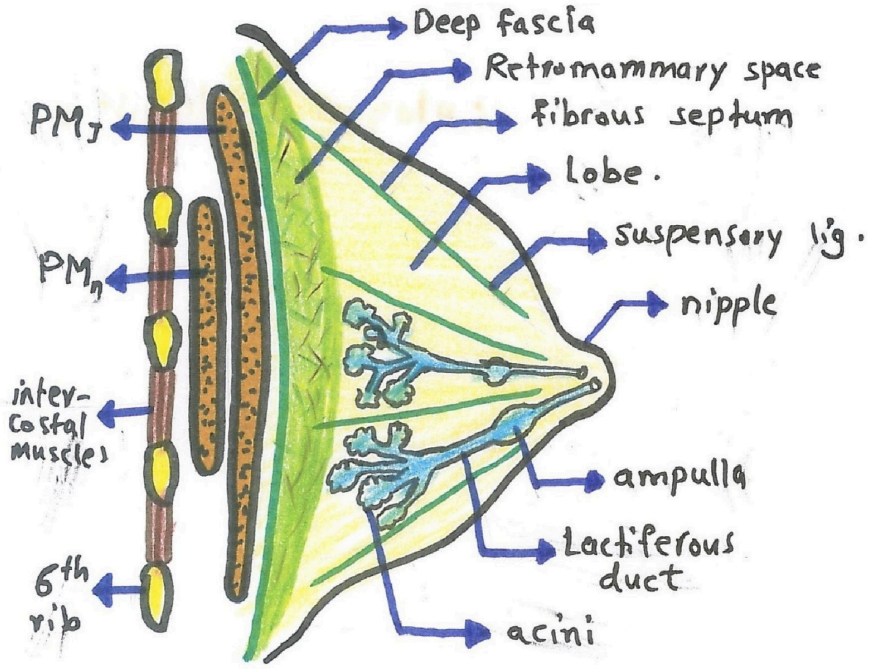
Shape & Position (cont.)



The bed of the breast with the pectoral fascia stripped away: pectoralis major above and serratus anterior inferolaterally, and the axillary process running up into the axilla.

Lobes, Ducts & the Retromammary Space

- Consists of 15–20 lobes separated by fibrous septa which is more developed in upper part near skin forming suspensory (Cooper's) ligament.
- Breast is separated from the deep fascia under it by a layer of loose areolar tissue called retromammary space (sub mammary space).
- Each lobe is formed of acini (lobules) (secreting milk) and transmits the milk by a lactiferous duct that opens into the nipple and before its end forms lactiferous sinus (ampulla), each lobe has one duct so there is 15–20 ducts in breast.



Dr. Milad's own sagittal drawing: deep fascia, retromammary space, fibrous septum, lobe, suspensory ligament, ampulla, lactiferous duct and acini.

Lobes, Ducts & the Retromammary Space (cont.)

N.B.

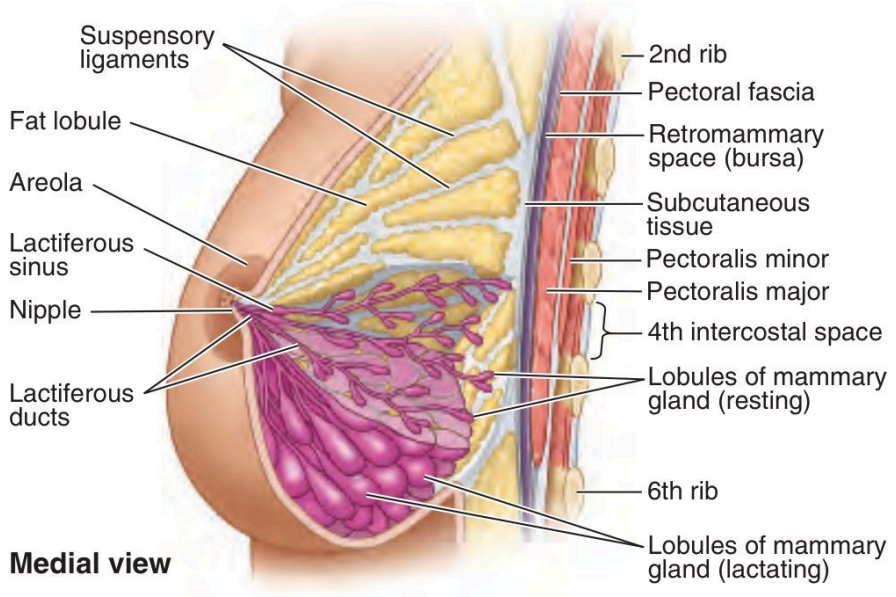
About 10 to 100 alveoli form one lobule; 20 to 40 lobules form one lobe.

N.B.

The breast has no capsule but embedded in fatty tissue.

N.B.

Nipple is directed downward and outward.



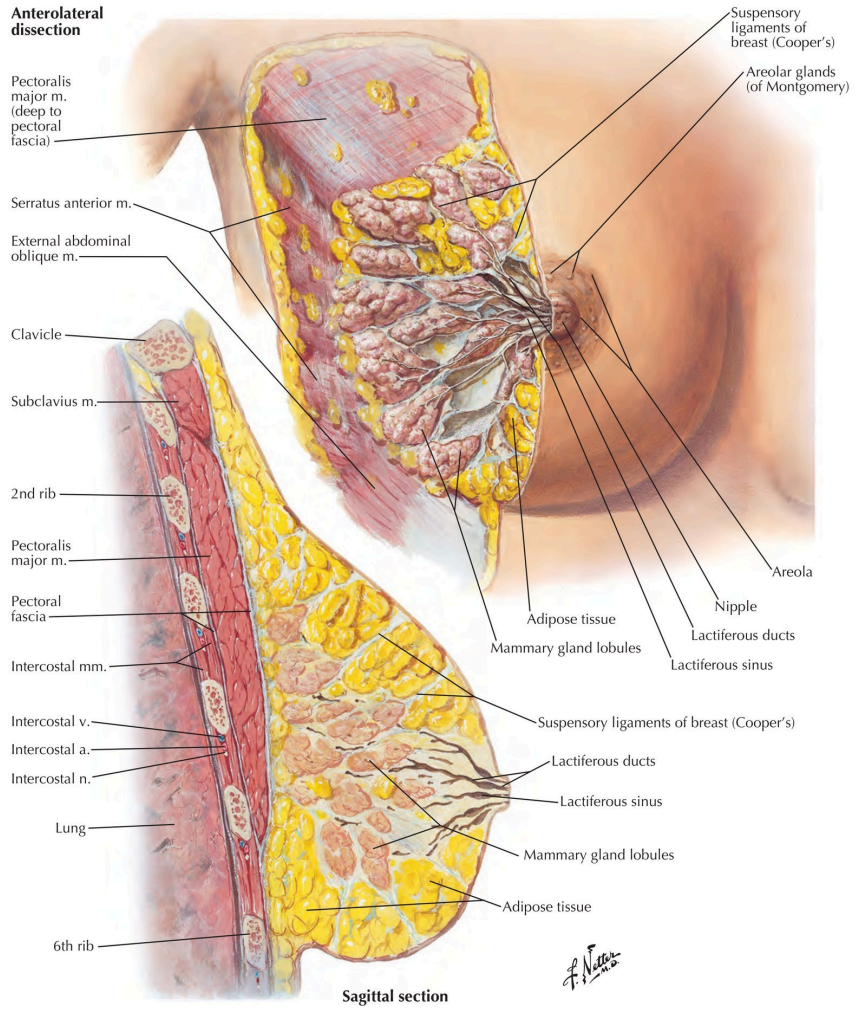
Sagittal section of the breast and anterior thoracic wall: the suspensory ligaments, the lactiferous ducts and sinus, and the retromammary space over the pectoral fascia.

Nipple & Areola

- Nipple: lies opposite to 4th intercostal space (4 inches from midline) but if large breast is lower, it has 15–20 opening.
- Areola: is a dark pink brownish circular area surrounding the nipple that change to dark brown by melanin at first pregnancy (for ever).
  - Areola contains sebaceous glands which is so large and visible forming cutaneous tubercles (Montgomery gland).
- Both nipple & areola have no fat under its skin.

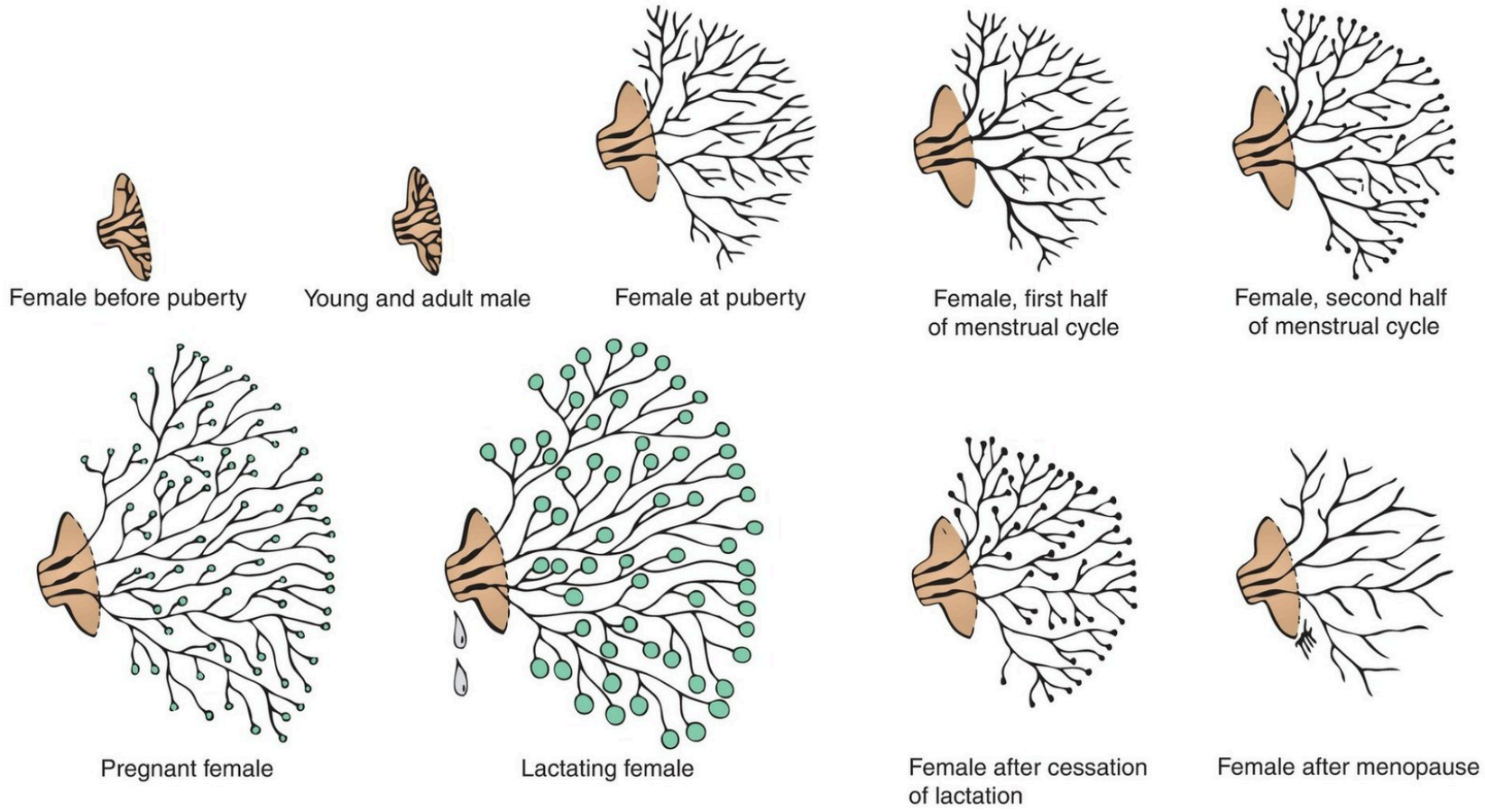
N.B.

The lactiferous ducts radiate outwards from the nipple, and so incision should be made in a radial direction to avoid cutting across the ducts. Axillary tail reach up to 3rd rib.



The mammary gland: the skin peeled back to show the lobes and lactiferous ducts converging on the nipple, the lactiferous sinus and the areolar glands of Montgomery.

Nipple & Areola (cont.)



The extent of the ducts and secretory alveoli in both sexes at different stages: before puberty, the young and adult male, puberty, the menstrual cycle, pregnancy, lactation and after the menopause.

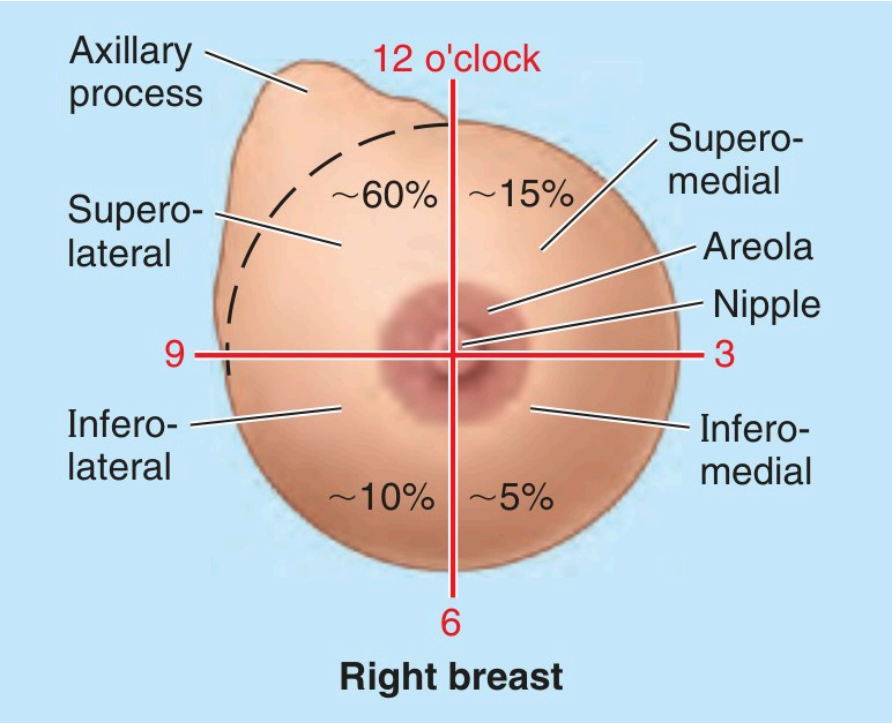
CHAPTER 2

Lymph Drainage of the Breast

A: Lymph from Skin (except nipple and areola)

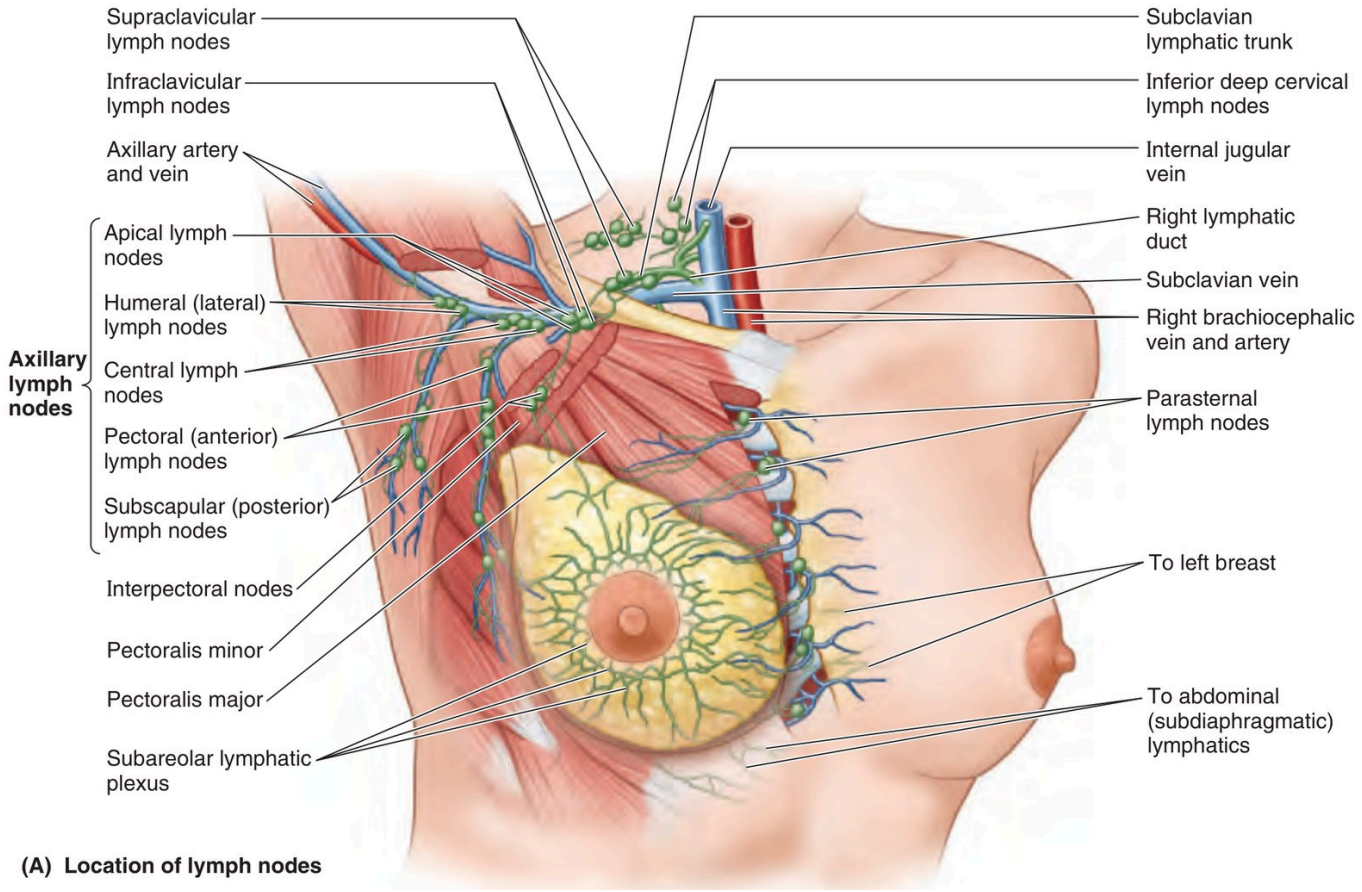
- Lateral quadrants: drain into anterior (pectoral) group (and lateral axillary LN).
- Upper quadrants: drain into apical group of lymph nodes.
- Lower quadrants: into lymph plexus of rectus sheath & subphrenic plexus.
- Medial quadrants: into internal thoracic (parasternal) nodes.

Some lymphatics of medial part across the front of sternum to anastomose with opposite breast (intermammary lymphatics).



The four quadrants of the breast and the axillary process, the map the drainage of the skin is described on.

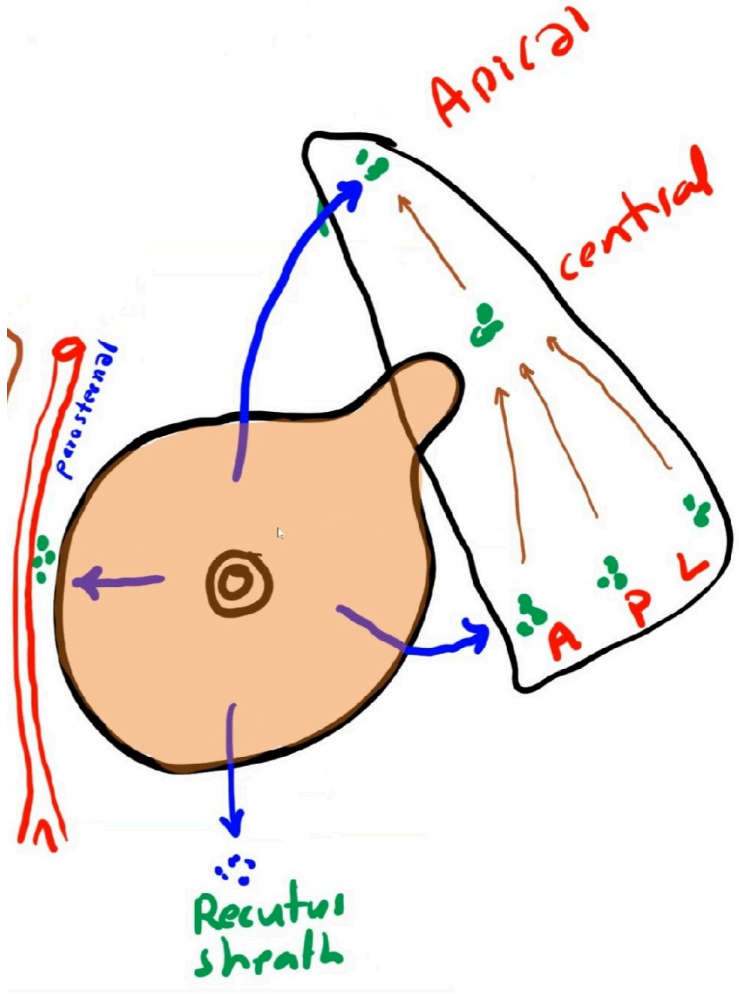
A: Lymph from Skin (except nipple and areola) (cont.)



Location of the lymph nodes of the breast, with the subareolar plexus, the pectoral and parasternal groups and the routes to the opposite breast and to the abdomen.

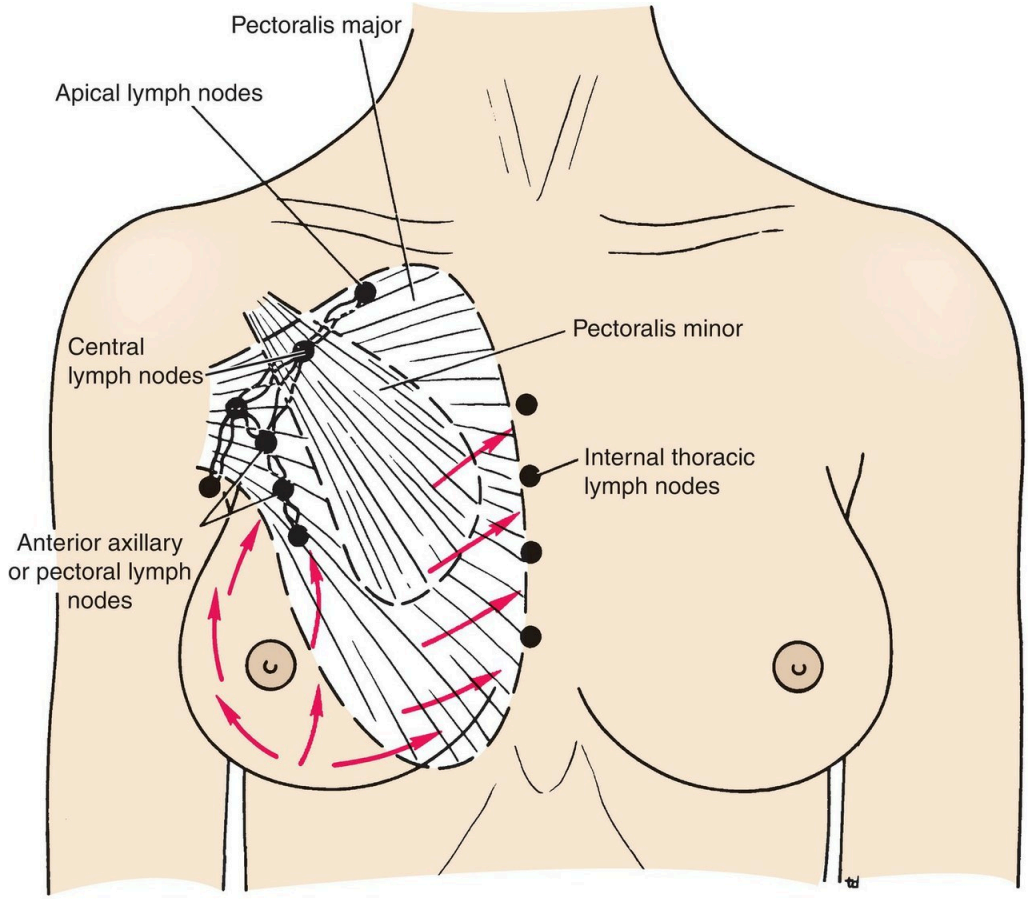
B: Lymph from Breast Tissue (including nipple and areola)

- 75 % to axillary LN.
- 20 % to parasternal LN (internal thoracic LN) — 25 %.
- 5 % to paravertebral LN (post. IC LN) — few.



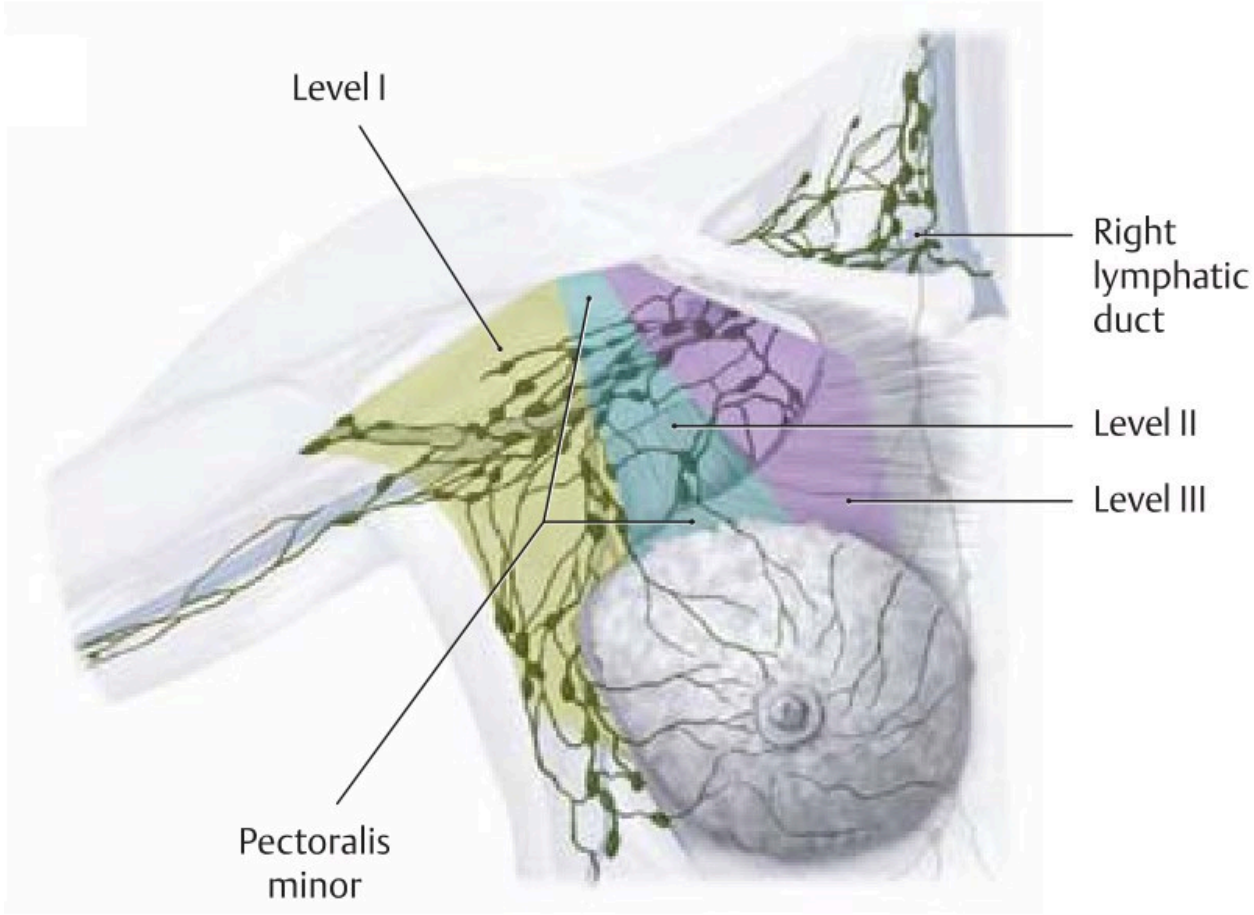
Dr. Milad's own drawing of the drainage: 75% laterally to the axillary nodes, 20% medially to the parasternal nodes and 5% back to the paravertebral nodes.

B: Lymph from Breast Tissue (including nipple and areola) (cont.)



The two routes: laterally to the anterior axillary (pectoral) nodes, and medially through the intercostal spaces to the internal thoracic (parasternal) nodes.

B: Lymph from Breast Tissue (including nipple and areola) (cont.)



The three levels of the axillary lymph nodes in relation to pectoralis minor.

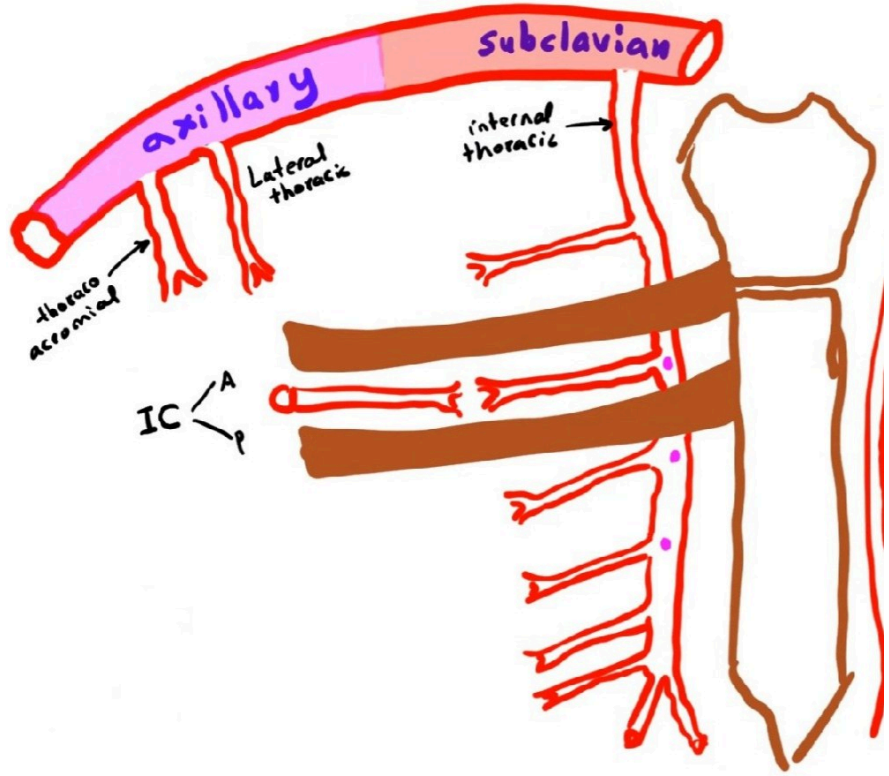
CHAPTER 3

The Blood Supply of the Breast

Blood Supply

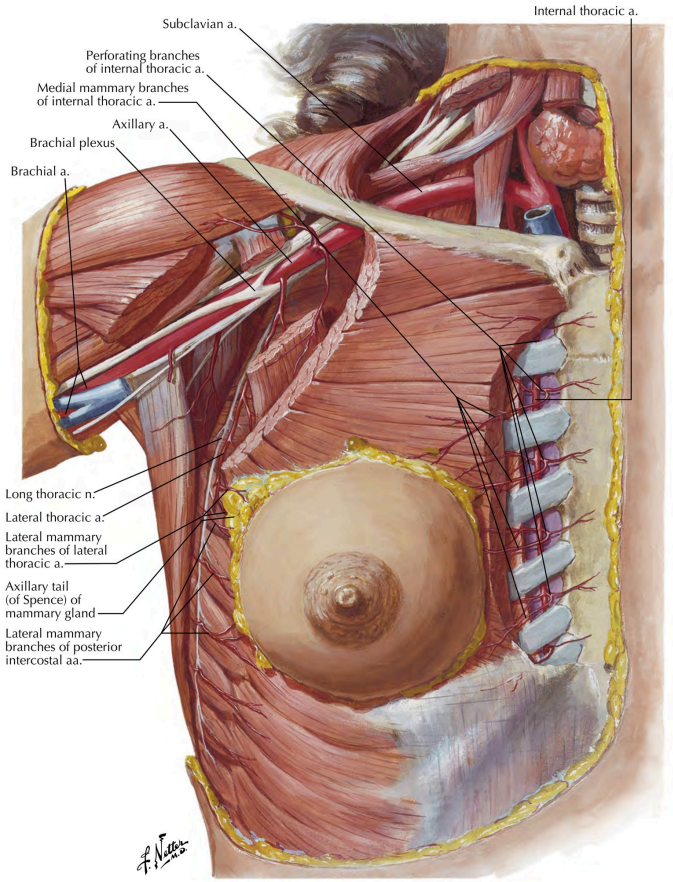
1. Perforating branches of internal thoracic artery.
2. Intercostal arteries (anterior and posterior IC) → from second to fifth IC arteries.
3. Lateral thoracic artery.
4. Thoracoacromial artery.

- Veins corresponding to arteries.



Dr. Milad's own drawing of the arterial supply: the internal thoracic and its perforating branches, the lateral thoracic and thoracoacromial arteries, and the intercostal arteries.

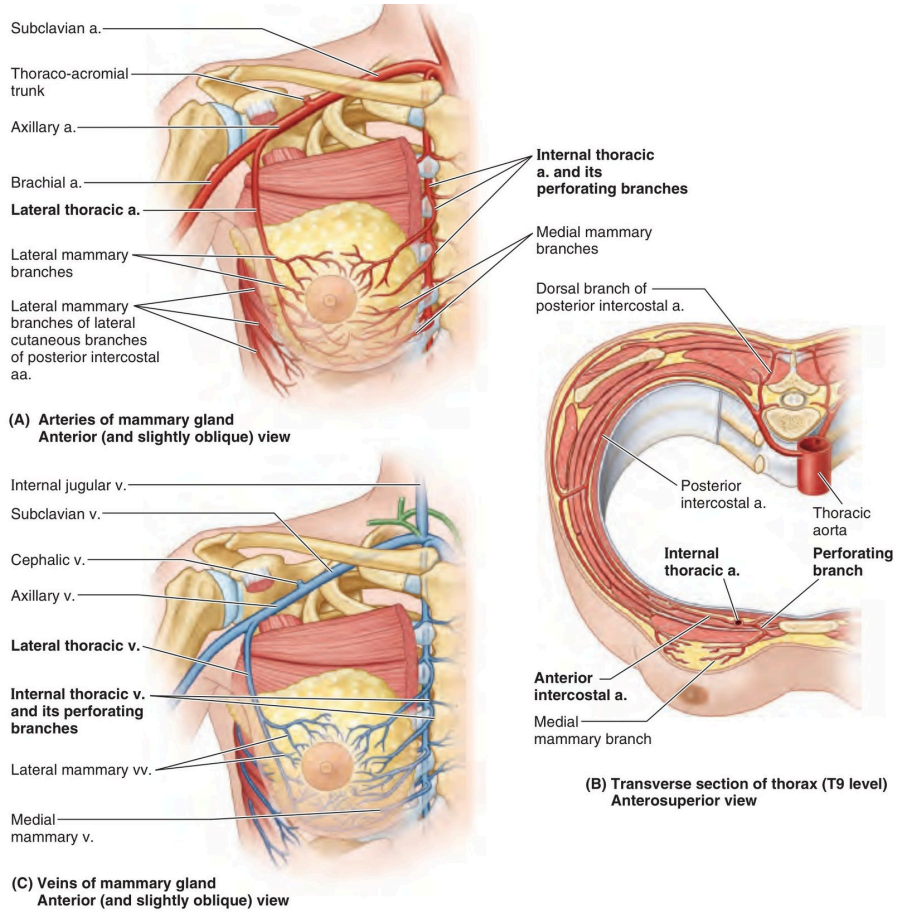
Blood Supply (cont.)



Arteries of the breast: the perforating and medial mammary branches of the internal thoracic, the lateral thoracic and its lateral mammary branches, and the posterior intercostals.

The Arteries of the Breast — Source & Share

Name of artery	Branched from	Part and % it supplies
Internal thoracic (mammary) artery	Subclavian artery	Medial part (30%)
Lateral thoracic (mammary) artery	Axillary artery	Lateral part (60%)
Intercostal artery	Aorta	Lower lateral quadrant (5–10%)
Thoracoacromial and superior thoracic arteries	Axillary artery	Upper part (1–5%)



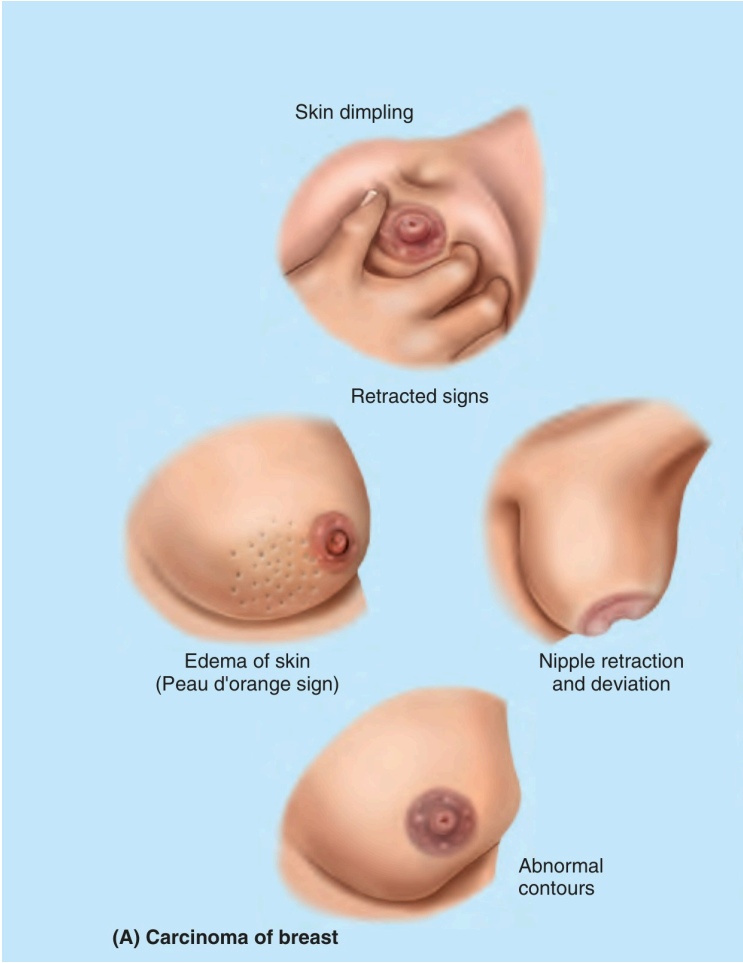
Vasculature of the breast: A, the arteries; B, a transverse section showing how the perforating and intercostal branches reach the gland; C, the corresponding veins.

CHAPTER 4

Clinical Notes

Clinical Notes

- Male breast may enlarge in some disease → "gynaecomastia".
- Suspensory (Cooper's) ligament may be invaded by breast cancer causing dimpling of skin "peau d'orange" by their shrinkage.
- Breast cancer may invade retromammary space into the muscle causing fixation of breast.
- Breasts at puberty assume their hemispherical shape under the influence of ovarian hormones & size ↑ by fat deposition.



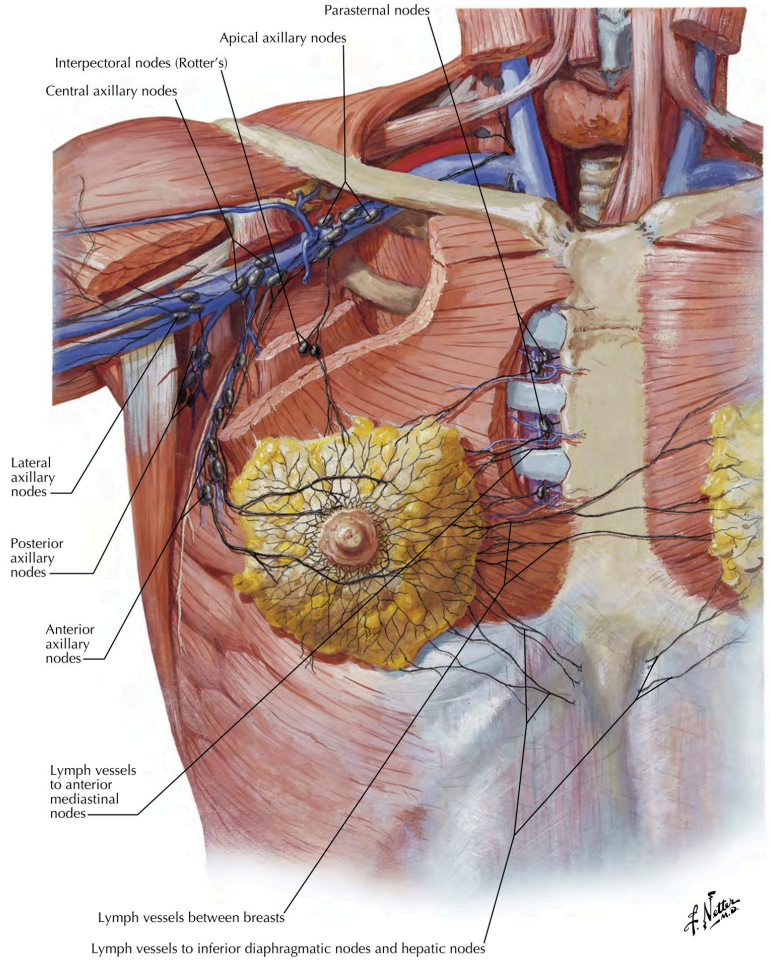
Carcinoma of the breast: skin dimpling and retraction from shrinkage of the suspensory ligaments, oedema of the skin (peau d'orange), nipple retraction and deviation, and abnormal contours.

Notes about the Breast

- Breast of male and immature female is similar in structure.
- The male breast consists of a system of ducts embedded in connective tissue that does not extend beyond margin of areola.
- Interlobular plexus lies between the lobules, sub-areolar plexus lies beneath areola.
- Few lymph vessels follow posterior intercostal arteries to drain into posterior intercostal lymph nodes (lateral quadrants).
- Mastitis: inflammation of breast.
- Poland syndrome: absent pectoralis major and/or breast.

N.B.

Connected to vertebral venous plexus by valveless veins so cancer breast can spread to vertebrae.

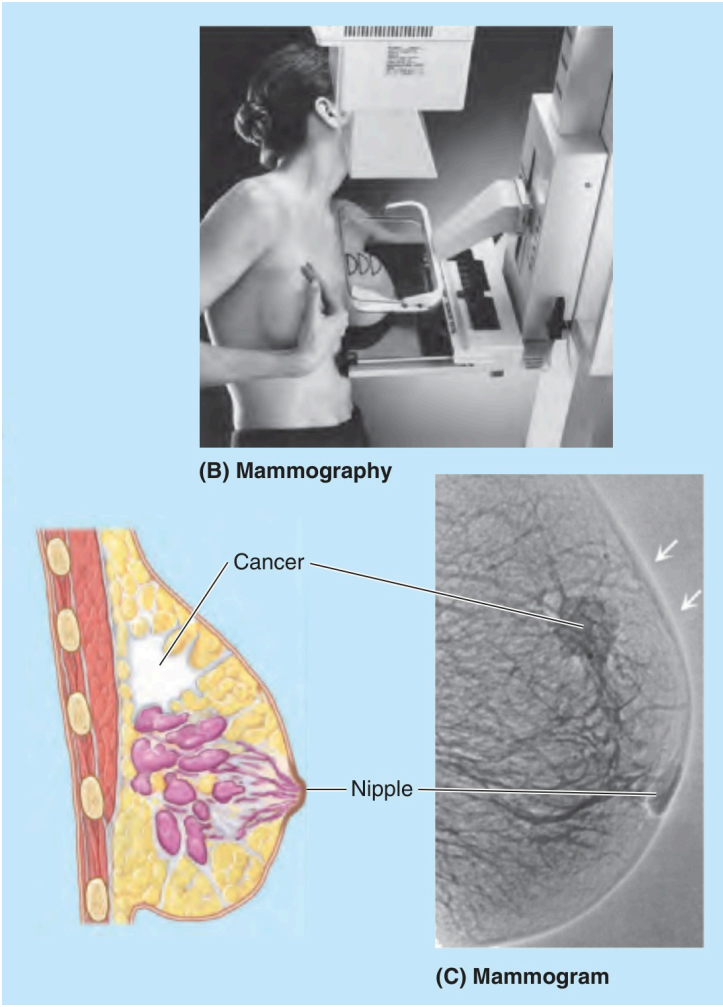


Lymph vessels and nodes of the breast: the anterior axillary (pectoral) group and the parasternal chain, with the subareolar plexus over the gland.

Notes about the Breast (cont.)

N.B.

Breast mammogram is used for screening of tumours.



Mammography, and a mammogram showing a carcinoma; beside it a section with the tumour lying in the substance of the gland.